



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Instrumentation Engineering
In-Semester Examination II
Academic Year: 2025-26 (Semester – I)

Class: SY	Subject: Sensors and Transducers	Subject Code: MDM-IN201
Date: 16.10.2025	Time: 14:30 to 15:15	Max. Marks: 20

Note:

- i. ATTEMPT ALL QUESTIONS. Draw neat diagrams wherever necessary.
- ii. Assume suitable data whenever necessary and state it clearly.
- iii. Mobile phones and programmable calculators are strictly prohibited.
- iv. Exchange/sharing of stationery, calculator, etc. is not allowed.
- v. Figures to the right indicate full marks.
- vi. The symbols carry the usual meaning.

Q.N.			Marks	CO	BT
Q.1	(a)	What are the basic pressure sensing elements? Draw three arrangements sensors based on resistive, inductive and capacitive principle using these basic pressure sensing elements for the measurement of force / pressure.	04	CO1 CO3	BT1 BT2
	(b)	What is the principle of capacitive transducer? Consider a non-conducting liquid in a tank and develop level measurement system of it using a capacitive transducer. A Capacitive transducer uses two quartz diaphragms of area 75mm ² separated by a distance of 3.5mm. A pressure of 900kN/m ² when applied to the top diaphragm produces a deflection of 0.6mm. The capacitance is 370pF when no pressure is applied to the diaphragms. Evaluate the value of capacitance after application of pressure 900kN/ m ² .	06	CO3 CO4	BT2 BT3 BT5
Q.2	(a)	What is an LVDT? What are the parameters that can be measured by this? Describe with neat diagram and output characteristics the principle of its construction and operation.	04	CO2	BT2
	(b)	Explain the construction and working principle of piezoelectric sensor? What is the difference between Piezo-electric and piezo-resistive sensors? A Quartz piezo-electric Crystal having a thickness of 2 mm and voltage sensitivity of 0.055 V-m/N is subjected to a pressure of 1.5 MN/m ² . Evaluate the voltage output.	06	CO3	BT4 BT5

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create Course

Outcomes:

	On successful completion of this course, students should be able to:
CO 1	To identify and define verity of sensors, signal conditioning devices and transducers.
CO 2	classify and explain with examples of transducer such as for measurement of temperature, strain, motion, position etc.
CO 3	choose proper sensor comparing different standards and guidelines to make sensitive measurements of physical parameters like pressure, flow, level, acceleration, etc.
CO 4	To analyse the problem using basic principles for development of project for agriculture, biomedical, Automobiles, Environmental, Petrochemical or other process industries.
CO 5	locate different type of sensors used in real life applications and paraphrase their importance
CO 6	To design, formulate, generate and deliver the solutions for given applications using best applicable sensors and transducers.



S. G. G. S. Institute of Engineering and Technology, Vishnupuri, Nanded.
ISE-II Examination Sem-I, AY 2025-26
Class: S.Y. BTech (Instrumentation Engineering)

Date: 14/10/2025
Time: 14.30 to 15.15 pm

Subject: SENSORS AND TRANSDUCERS
Code: PCC-IN201

Max. Marks: 20

Sr. No	Questions	Marks	COs	BT
Q. 1	A. What do you understand by passive transducer? Give examples of passive transducer.	(01)	CO1 ,CO2	BT1
	B. Solve following problems (any TWO):	(04)	CO1	BTV
	I. An LVDT produces rms output voltage of 2.6 V for displacement of 0.4 μm . Calculate the sensitivity of LVDT.			
	II. Strain gauge of 120 ohm nominal resistance is fixed on a structural member subjected to a strain of 50 $\mu\text{m}/\text{m}$. If the gauge factor is 2.5, what is the change in resistance of the gauge?			
Q. 2	III. A parallel plate capacitor transducer uses plates of area 500 mm^2 which are separated by distance 0.2 mm. Calculate the value of capacitance when the dielectric is air having a permittivity of $8.85 \times 10^{-12} \text{ F/m}$.			
	Define the gauge factor of a resistance strain gauge and obtain the expression for the voltage sensitive Full bridge. What are the advantages of this configuration?	(05)	CO2	BT1
Q.3	Describe the constructional features of a linear variable differential transformer. Give its merits and demerits.	(05)	CO2	BT1
Q.4	A. Show how a capacitive transducer can be used to monitor the thickness of an insulating sheet in motion without making physical contact.	(05)	CO4	BT1

Bloom's taxonomy: BT1: Remember BT2: Understand BT3: Apply BT4: Analyse BT5 Evaluate
BT6 Create

Course outcome:

1. To identify, list, define verity of sensors, signal conditioning devices, transducer (Primary and secondary).
2. To describe, draw, classify and produced sketches, drawings to explain working principles of various sensors and transducers.
3. To analyze the problem using basic principles for development of project for agriculture, biomedical, Automobiles, Environmental, Petrochemical or other process industries.
4. To monitor and evaluate asses and compare of various sensors and transducers and came to conclusion for the best selection for the desired applications.
5. To create, design, formulate, generate and deliver the solutions for given applications using best applicable sensors and transducers.